MA3236 NONLINEAR PROGRAMMING
Semester 1, AY 2026/2027
Tutorial 7

Review: Lecture 14 — convex KKT and SVM

Consider

$$\min_{\mathbf{x}\in\mathcal{D}} f(\mathbf{x}) \quad \text{s.t.} \quad \mathbf{g}(\mathbf{x}) = \mathbf{0}, \qquad \mathbf{h}(\mathbf{x}) \le \mathbf{0},$$

where $\mathcal{D}$ is open convex, $f, h_j \in C^1$ are convex, and g_i are affine. Write $\mathcal{C}$ for the feasible set.

Certifying a candidate. For a feasible $\mathbf{x}^*$, use $L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f(\mathbf{x}) + \boldsymbol{\lambda}^T\mathbf{g}(\mathbf{x}) + \boldsymbol{\nu}^T\mathbf{h}(\mathbf{x})$. Seek free equality multipliers $\boldsymbol{\lambda}^*$ and inequality multipliers $\boldsymbol{\nu}^*$ satisfying

$$\nabla_{\mathbf{x}} L(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) = \mathbf{0}, \qquad \boldsymbol{\nu}^* \ge \mathbf{0}, \qquad \nu_j^* h_j(\mathbf{x}^*) = 0 \quad (\forall j).$$

These are KKT, with feasibility already imposed. Convexity makes them sufficient for global optimality without a CQ. Conversely, LICQ at a minimizer gives KKT and unique multipliers. An active constraint may still have zero multiplier.

Separating existence from uniqueness.

- Existence from previous lectures: continuous f on nonempty compact $\mathcal{C}$ ensures attainment; strict convexity on $\mathcal{C}$ gives at most one minimizer, but not attainment.
- For convex C^2 data, KKT + SOSC also gives a unique global minimum. Strict convexity and SOSC do not imply each other in general; primal uniqueness need not give multiplier uniqueness.

Using the certificate for hard-margin SVM. For labeled data $(\mathbf{x}_i, y_i)$, $y_i \in \{-1, 1\}$, normalizing the margins to $\mathbf{w}^T\mathbf{x} + b = \pm 1$ gives width $2/\|\mathbf{w}\|$. Maximizing it gives

$$\min_{\mathbf{w},b} \tfrac{1}{2}\|\mathbf{w}\|^2 \quad \text{s.t.} \quad h_i(\mathbf{w}, b) = 1 - y_i(\mathbf{w}^T\mathbf{x}_i + b) \le 0 \quad (i = 1, \dots, N).$$

This is a convex program. Differentiating its Lagrangian in $\mathbf{w}, b$ gives the certificate

$$\mathbf{w} = \sum_i \nu_i y_i \mathbf{x}_i, \qquad \sum_i \nu_i y_i = 0, \qquad h_i \le 0, \quad \nu_i \ge 0, \quad \nu_i h_i = 0.$$

The separator is $\mathbf{w}^T\mathbf{x} + b = 0$; $\nu_i > 0$ identifies a loaded support vector on a margin (the converse can fail). Both classes and strict separability give a unique $(\mathbf{w}^*, b^*)$, but not necessarily unique multipliers.

Allowing margin violations. If hard-margin constraints are infeasible, introduce slacks $\xi_i \ge 0$ and penalize them with $C > 0$:

$$\min_{\mathbf{w},b,\boldsymbol{\xi}} \tfrac{1}{2}\|\mathbf{w}\|^2 + C\sum_i \xi_i \quad \text{s.t.} \quad 1 - y_i(\mathbf{w}^T\mathbf{x}_i + b) \le \xi_i, \quad \xi_i \ge 0.$$

At fixed $(\mathbf{w}, b)$, the smallest slack is $\xi_i^{\min} = [1 - y_i(\mathbf{w}^T\mathbf{x}_i + b)]_+$. Thus $\xi_i^{\min} = 0$ means on/outside the correct margin; $0 < \xi_i^{\min} < 1$ means inside it; $\xi_i^{\min} = 1$ means on the separator; and $\xi_i^{\min} > 1$ means on the wrong side. For finite data, the soft-margin model is convex, feasible, and attains a minimum.

Review: Lecture 15 — QP and ALM

For $\min f(\mathbf{x})$ subject to $\mathbf{g}(\mathbf{x}) = \mathbf{0}$, let $f, g_i \in C^2$ on open $\mathcal{D}$. Write

$$L = f + \boldsymbol{\lambda}^T \mathbf{g}, \qquad \mathbf{A} = [\nabla g_1, \ldots, \nabla g_m], \qquad \mathbf{H}_L = \nabla^2_{\mathbf{xx}} L.$$

Quadratic penalty method. The quadratic penalty (QP) method uses an effective multiplier ($\rho > 0$) to interpret its derivatives:

$$P_\rho = f + \tfrac{\rho}{2}\|\mathbf{g}\|^2, \qquad \widehat{\boldsymbol{\lambda}} = \rho \mathbf{g},$$
$$\nabla P_\rho = \nabla f + \mathbf{A}\widehat{\boldsymbol{\lambda}}, \quad \nabla^2 P_\rho = \mathbf{H}_L(\mathbf{x}, \widehat{\boldsymbol{\lambda}}) + \rho \mathbf{A}\mathbf{A}^T.$$

- Inner stationarity supplies KKT stationarity, but not feasibility. At a LICQ KKT pair with $\boldsymbol{\lambda}^* \neq \mathbf{0}$, no finite ρ gives $\nabla P_\rho(\mathbf{x}^*) = \mathbf{0}$. The traditional rule increases $\rho^{(k+1)} = \gamma \rho^{(k)}$ with fixed $\gamma > 1$.
- Large normal curvature with bounded tangent curvature can cause ill-conditioning.

Augmented Lagrangian method. The augmented Lagrangian method (ALM) adds a linear correction: with fixed $\rho > 0$, minimize $L_A(\cdot, \boldsymbol{\lambda}^{(k)}; \rho)$, then update:

$$L_A = f + \boldsymbol{\lambda}^T \mathbf{g} + \tfrac{\rho}{2}\|\mathbf{g}\|^2, \qquad \boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \rho \mathbf{g}(\mathbf{x}^{(k+1)}).$$

Substituting the update gives

$$\nabla_{\mathbf{x}} L_A(\mathbf{x}^{(k+1)}, \boldsymbol{\lambda}^{(k)}; \rho) = \nabla_{\mathbf{x}} L(\mathbf{x}^{(k+1)}, \boldsymbol{\lambda}^{(k+1)}).$$

Inner stationarity uses the updated multiplier, so feasibility can improve at fixed ρ.

Separate inner and outer tests. Warm-start SD or Newton with line search on $\Psi^{(k)} = P_{\rho^{(k)}}$ or $L_A(\cdot, \boldsymbol{\lambda}^{(k)}; \rho)$, until

$$\|\nabla \Psi^{(k)}(\mathbf{x}^{(k+1)})\|_\infty \leq \varepsilon_{\text{in}}^{(k)} < \varepsilon_{\text{out}}, \qquad \varepsilon_{\text{in}}^{(k)} > 0.$$

With the QP estimate or updated ALM multiplier, require both KKT residuals:

$$r_{\text{KKT}} = \max\{\|\nabla f + \mathbf{A}\boldsymbol{\lambda}\|_\infty, \|\mathbf{g}\|_\infty\} \leq \varepsilon_{\text{out}}.$$

A small inner gradient alone cannot certify feasibility. A fixed inner tolerance does not in general guarantee exact asymptotic convergence.

Match guarantees to assumptions.

- For QP with $\rho^{(k)} \to \infty$ and a nonempty feasible set, exact global inner minima give globally optimal cluster points in $\mathcal{D}$. For stationary solves, bounded $\widehat{\boldsymbol{\lambda}}$ forces $\mathbf{g} = \widehat{\boldsymbol{\lambda}}/\rho \to \mathbf{0}$; vanishing inner gradients and a convergent primal–multiplier subsequence in $\mathcal{D} \times \mathbf{R}^m$ give KKT. Local inner minima alone may remain infeasible.
- For ALM, KKT + LICQ + SOSC gives local exactness at the correct multiplier for sufficiently large finite ρ; local convergence uses the nearby minimizing branch.

Extend to inequalities. For $\mathbf{h} \leq \mathbf{0}$, use the following added terms and multipliers:

$$\text{QP}: \quad \tfrac{\rho}{2}\|[\mathbf{h}]_+\|^2, \qquad \widehat{\boldsymbol{\nu}} = \rho[\mathbf{h}]_+,$$
$$\text{ALM}: \quad \sum_j \frac{[\nu_j + \rho h_j]_+^2 - \nu_j^2}{2\rho}, \quad \nu_j^+ = [\nu_j + \rho h_j]_+.$$

Also test dual feasibility and complementarity.

1. **A convex KKT certificate with an active zero multiplier.** Consider

$$\min_{\mathbf{x}\in\mathbf{R}^3} \frac{1}{2}\|\mathbf{x}-\mathbf{c}\|^2 \quad \text{s.t.} \quad \mathbf{e}^T\mathbf{x}=4, \qquad \mathbf{x}\geq\mathbf{0},$$

where $\mathbf{c}=(4,2,1)^T$ and $\mathbf{e}=(1,1,1)^T$.

(a) Show that a unique global minimizer exists and that LICQ holds at every feasible point.

(b) Write and solve the KKT conditions, using $h_i(\mathbf{x})=-x_i$. Find all multipliers and the active set. Does strict complementarity hold?

Solution. The simplex is nonempty (for example, $(4,0,0)^T$ is feasible), closed and bounded. Continuity gives attainment, and strict convexity gives uniqueness. For LICQ, let J be the zero coordinates of a feasible point. At least one coordinate ℓ is positive. If $a\mathbf{e}-\sum_{j\in J} b_j\mathbf{e}_j=\mathbf{0}$, the ℓth coordinate gives $a=0$, and the other coordinates then give every $b_j=0$. Thus the equality gradient and active inequality gradients are independent.

Stationarity, feasibility and complementarity give

$$\mathbf{x}-\mathbf{c}+\lambda\mathbf{e}-\boldsymbol{\nu}=\mathbf{0}, \qquad \mathbf{e}^T\mathbf{x}=4, \quad \mathbf{x}\geq\mathbf{0}, \quad \boldsymbol{\nu}\geq\mathbf{0}, \quad \nu_i x_i=0.$$

If $x_i>0$, then $\nu_i=0$ and $x_i=c_i-\lambda$. If $x_i=0$, then $\nu_i=\lambda-c_i\geq 0$. Hence $x_i=\max\{c_i-\lambda,0\}$, and the sum constraint becomes

$$\max\{4-\lambda,0\}+\max\{2-\lambda,0\}+\max\{1-\lambda,0\}=4.$$

It is solved by $\lambda^*=1$, giving

$$\mathbf{x}^*=(3,1,0)^T, \qquad \boldsymbol{\nu}^*=(0,0,0)^T.$$

All KKT conditions hold, so convex KKT sufficiency certifies global optimality. LICQ makes these multipliers unique. The active set is $J=\{3\}$, but $\nu_3^*=0$, so strict complementarity fails.

2. **Hard and soft margins.** The observations are

$$\mathbf{x}_1=(-2,0)^T,\ y_1=-1; \qquad \mathbf{x}_2=(1,0)^T,\ y_2=1; \qquad \mathbf{x}_3=(3,1)^T,\ y_3=1.$$

(a) Write the hard-margin SVM and its complete KKT conditions.

(b) Use the first two constraints to obtain a lower bound on w_1. Find an optimal classifier and a multiplier certificate, and report its separator, margin width and loaded support vectors.

(c) Add a fourth observation $\mathbf{x}_4 = (1,0)^T$, $y_4 = -1$. Explain why hard margin is now infeasible. Write the soft-margin model with $C > 0$ and give one feasible point; you need not solve this model.

Solution. For the original three observations, the objective is $\frac{1}{2}(w_1^2 + w_2^2)$ and the inequalities are

$$h_1 = 1 - 2w_1 + b \le 0, \qquad h_2 = 1 - w_1 - b \le 0, \qquad h_3 = 1 - 3w_1 - w_2 - b \le 0.$$

The remaining KKT conditions are $\nu_i \ge 0$, $\nu_i h_i = 0$, and

$$w_1 = 2\nu_1 + \nu_2 + 3\nu_3, \qquad w_2 = \nu_3, \qquad -\nu_1 + \nu_2 + \nu_3 = 0.$$

The first two primal inequalities require $1 - w_1 \le b \le 2w_1 - 1$, hence $w_1 \ge 2/3$. Therefore every feasible objective is at least $2/9$. The choices

$$\mathbf{w}^* = (2/3, 0)^T, \quad b^* = 1/3, \qquad \nu_1^* = \nu_2^* = 2/9, \quad \nu_3^* = 0$$

attain that bound and satisfy every KKT equation: $h_1 = h_2 = 0$, $h_3 = -4/3 < 0$, and the displayed stationarity equations hold. Equality in the objective bound forces $w_1 = 2/3, w_2 = 0$, and the interval for b then forces $b = 1/3$, proving uniqueness. The separator is $x_1 = -1/2$, the margin width is 3, and the loaded support vectors are observations $1, 2$.

With the fourth observation, the same score $s = w_1 + b$ would have to satisfy $s \ge 1$ for observation 2 and $s \le -1$ for observation 4. This is impossible. The soft-margin problem is

$$\begin{aligned} \min_{\mathbf{w},b,\boldsymbol{\xi}} \quad & \tfrac{1}{2}(w_1^2 + w_2^2) + C\sum_{i=1}^{4} \xi_i \\ \text{s.t.} \quad & 1 - y_i(\mathbf{w}^T\mathbf{x}_i + b) - \xi_i \le 0, \quad -\xi_i \le 0, \qquad i = 1, \dots, 4. \end{aligned}$$

For example, $\mathbf{w} = \mathbf{0}, b = 0, \xi_i = 1$ is feasible. Its objective is convex and all constraints are affine.

3. **A QP path and its cost.** Consider the equality-constrained problem

$$\min_{\mathbf{x} \in \mathbf{R}^2} f(\mathbf{x}) = x_1^2 + x_2^2 \quad \text{s.t.} \quad g(\mathbf{x}) = x_1 + x_2 + 1 = 0.$$

(a) Find the constrained minimizer and its multiplier. For $P_\rho = f + \rho g^2/2$ ($\rho \ge 0$), find the exact inner minimizer $\mathbf{x}(\rho)$, the estimate $\widehat{\lambda}(\rho) = \rho g(\mathbf{x}(\rho))$, and their limits as $\rho \to \infty$.

(b) Find the eigenvalues and condition number of $\nabla^2 P_\rho$. At $\mathbf{x}(\rho)$ compute $r_{\text{stat}}, r_{\text{feas}}, r_{\text{KKT}}$ using the estimated multiplier. How large must ρ be to attain $\varepsilon_{\text{out}} = 10^{-3}$? Explain why exact inner stationarity alone is an incomplete stopping test.

Solution. Constrained stationarity gives $2x_1 + \lambda = 2x_2 + \lambda = 0$. Combining this with $x_1 + x_2 = -1$ yields

$$\mathbf{x}^* = (-1/2, -1/2)^T, \qquad \lambda^* = 1.$$

Strict convexity on the affine feasible set certifies the unique minimum. For the penalty subproblem, subtracting its two gradient equations gives $x_1 = x_2 = t$. Then

$$2t + \rho(2t + 1) = 0, \qquad t = -\frac{\rho}{2(1+\rho)}.$$

The positive-definite Hessian makes this the unique global inner minimizer. Thus

$$\mathbf{x}(\rho) = -\frac{\rho}{2(1+\rho)}\begin{pmatrix}1\\1\end{pmatrix}, \qquad g(\mathbf{x}(\rho)) = \frac{1}{1+\rho}, \qquad \widehat{\lambda}(\rho) = \frac{\rho}{1+\rho}.$$

The point tends to $\mathbf{x}^*$ and the multiplier tends to 1. For every finite $\rho > 0$, the inner minimizer remains infeasible.

Since

$$\nabla^2 P_\rho = 2I + \rho\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix},$$

the tangent vector $(1, -1)^T$ has eigenvalue 2 and the normal vector $(1, 1)^T$ has eigenvalue $2 + 2\rho$. Its spectral condition number is $1 + \rho$. The residuals are

$$r_{\text{stat}} = 0, \qquad r_{\text{feas}} = r_{\text{KKT}} = \frac{1}{1+\rho}.$$

The requested outer accuracy requires $\rho \geq 999$. The inner gradient vanishes at every one of these points, including those with large feasibility violation. Increasing the penalty controls that violation but makes the inner quadratic more poorly conditioned.

4. **Fixed-penalty ALM.** Use the problem in Question 3 and $L_A(\mathbf{x}, \lambda; \rho) = f(\mathbf{x}) + \lambda g(\mathbf{x}) + \rho g(\mathbf{x})^2/2$. Throughout this question $\rho > 0$ is fixed.

   (a) With exact inner minimization, derive the returned point $\mathbf{x}^{(k+1)}$ and the multiplier recurrence. Show that the multiplier error contracts and that $\{\mathbf{x}^{(k)}\}$ converges to the constrained solution.

   (b) Set $\rho = 1$, $\lambda^{(0)} = 0$. Compute the first two returned points and updated multipliers. Compare feasibility and inner Hessian conditioning with Question 3.

Solution. The two inner stationarity equations again give $x_1 = x_2 = t$, now with

$$2t + \lambda^{(k)} + \rho(2t + 1) = 0.$$

Hence

$$\mathbf{x}^{(k+1)} = -\frac{\lambda^{(k)} + \rho}{2(1+\rho)} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad g(\mathbf{x}^{(k+1)}) = \frac{1 - \lambda^{(k)}}{1+\rho}.$$

Substitute the latter expression into the multiplier update:

$$\lambda^{(k+1)} = \lambda^{(k)} + \frac{\rho(1 - \lambda^{(k)})}{1+\rho} = \frac{\lambda^{(k)} + \rho}{1+\rho}.$$

Subtracting the fixed point $\lambda^* = 1$ gives

$$\lambda^{(k+1)} - 1 = \frac{\lambda^{(k)} - 1}{1+\rho}.$$

Since $1/(1+\rho) < 1$, the multiplier converges to 1 and the formula for the primal point gives $\mathbf{x}^{(k+1)} \to (-1/2, -1/2)^T$.

For $\rho = 1$ and $\lambda^{(0)} = 0$,

$$\mathbf{x}^{(1)} = (-1/4, -1/4)^T, \quad \lambda^{(1)} = 1/2, \qquad \mathbf{x}^{(2)} = (-3/8, -3/8)^T, \quad \lambda^{(2)} = 3/4.$$

Their feasibility residuals are $1/2$ and $1/4$, respectively; stationarity with the updated multiplier is zero. The inner Hessian is always the matrix from Question 3 with $\rho = 1$, so its condition number stays at 2. Feasibility improves through the multiplier corrections without increasing this condition number.